

Lower bounds for perfect difference sets, infinitely often

Erdős Problem #1194 — expository note

April 30, 2026

Abstract

We give two self-contained derivations of i.o. lower bounds on perfect difference sets. Theorem 2 reproves the classical $a_n \gg n \log n$ infinitely often via Sidon density (Erdős, [1]). Theorem 3 recovers the recent refinement $a_n \gg n^{2-o(1)}$ infinitely often, due to GPT-5.4 Pro and streamlined by Bloom in the comment thread of erdosproblems.com/1194 (April 23 2026). The latter argument does not use Sidon density at all; it uses only the PDS partition identity and a gap lower bound from the hypothetical pointwise upper bound on a_n . We also flag a typographical error on the live problem page (“diverges” vs “converges”).

1 Setup and notation

A set $A \subset \mathbb{N}$ is a *perfect difference set* (PDS) if every $n \geq 1$ admits a *unique* representation

$$n = a_n - b_n, \quad a_n, b_n \in A, \quad a_n > b_n.$$

The notation a_n is fixed by this definition: it is the larger element of the unique representing pair for n . It is *not* the n -th element of A . To avoid confusion, we write the increasing enumeration of A as

$$A = \{x_1 < x_2 < x_3 < \cdots\},$$

and use a_n exclusively for the representing pair.

A PDS is in particular a Sidon set: all pairwise differences are distinct. We further write $A_x := A \cap [1, x]$, $k(x) := |A_x|$, and $N(x) := \#\{n \geq 1 : a_n \leq x\}$.

2 The PDS counting identity

Lemma 1 (PDS counting). *For every $x \geq 1$,*

$$N(x) = \binom{k(x)}{2}.$$

Proof. Each $n \leq N(x)$ has $a_n \leq x$ and so $b_n < a_n \leq x$, putting both elements of the unique representing pair in A_x . The map $n \mapsto \{a_n, b_n\}$ is therefore an injection from $\{n : a_n \leq x\}$ into the unordered pairs of A_x , so $N(x) \leq \binom{k(x)}{2}$. Conversely, every unordered pair $\{a, b\} \subset A_x$ with $a > b$ produces a difference $a - b \in [1, x - 1]$, which by uniqueness is the representation of some unique n with $a_n = a$, $b_n = b$, and $a_n \leq x$. Hence $\binom{k(x)}{2} \leq N(x)$. $\square$

Corollary 1. $\binom{k(x)}{2} \leq x$ and so $k(x) \leq (1 + o(1))\sqrt{2x}$.

This recovers the standard Sidon density bound; the PDS gives *equality*, not only inequality.

3 Classical $n \log n$ infinitely often

Theorem 1 (Erdős, after [1]). *For every infinite Sidon set $B \subset \mathbb{N}$,*

$$\liminf_{x \rightarrow \infty} \frac{|B \cap [1, x]|}{\sqrt{x/\log x}} < \infty.$$

We use Theorem 1 as a black box; the proof is elementary and standard.

Theorem 2. *For any perfect difference set, $a_n \gg n \log n$ for infinitely many n .*

Proof. Choose $x_j \rightarrow \infty$ with $k(x_j) \leq C\sqrt{x_j/\log x_j}$ by Theorem 1. By Lemma 1,

$$N(x_j) = \binom{k(x_j)}{2} \leq \frac{C^2 x_j}{2 \log x_j}.$$

Setting $n_j := N(x_j) + 1$ gives $a_{n_j} > x_j$ and $n_j \leq C^2 x_j / (2 \log x_j) + 1$. Hence

$$a_{n_j} > x_j \geq \frac{2 \log x_j}{C^2} (n_j - 1) \geq \frac{2(1 + o(1))}{C^2} n_j \log n_j,$$

using $\log x_j \geq (1 + o(1)) \log a_{n_j} \geq (1 + o(1)) \log n_j$. $\square$

4 The refinement to $n^{2-o(1)}$

The refinement that follows is due to GPT-5.4 Pro and was condensed by Thomas Bloom in the comment thread on the problem page. It does not use Sidon density at all. The argument hinges on two structural facts about PDS that we extract first.

4.1 Two preparatory lemmas

For each $k \geq 2$, let

$$D_k := \{x_k - x_i : 1 \leq i < k\}.$$

Lemma 2 (PDS partition). *The family $\{D_k\}_{k \geq 2}$ is a partition of $\mathbb{Z}_{>0}$. In particular, for every $X \geq 1$,*

$$X = \sum_{k \geq 2} |D_k \cap [1, X]|.$$

Proof. Every $n \geq 1$ has a unique representation $n = a_n - b_n$ with $a_n, b_n \in A$, $a_n > b_n$. Pick k such that $a_n = x_k$ and let i be such that $b_n = x_i$; then $i < k$ and $n = x_k - x_i \in D_k$. Conversely, every element $x_k - x_i \in D_k$ is a positive integer. Uniqueness in PDS gives that n lies in exactly one D_k . $\square$

Lemma 3 (Gap lower bound from $a_n \ll n^c$). *Suppose $a_n \leq C n^c$ for all $n \geq 1$ and some $c > 1$. Then for every $k \geq 2$,*

$$x_k - x_{k-1} \geq (x_k/C)^{1/c}.$$

Consequently, $x_k \gg k^{c/(c-1)}$.

Proof. Let $d_k := x_k - x_{k-1}$. The pair (x_k, x_{k-1}) realises the difference d_k ; by PDS uniqueness, $a_{d_k} = x_k$. The hypothesis gives $x_k = a_{d_k} \leq C d_k^c$, so $d_k \geq (x_k/C)^{1/c}$. Telescoping,

$$x_k \geq x_k - x_1 = \sum_{i=2}^k d_i \geq C^{-1/c} \sum_{i=2}^k x_i^{1/c}.$$

Standard induction (or comparison with the ODE $y' = C^{-1/c} y^{1/c}$) yields $x_k \gg k^{c/(c-1)}$. $\square$

4.2 Main contradiction

Theorem 3 (GPT-5.4 Pro, after Bloom's exposition). *For any perfect difference set and any $c < 2$, the inequality $a_n \leq Cn^c$ cannot hold for all n and any constant C . Equivalently, for every $c < 2$,*

$$\limsup_{n \rightarrow \infty} \frac{a_n}{n^c} = \infty,$$

i.e. $a_n \gg n^{2-o(1)}$ infinitely often.

Proof. Suppose, for contradiction, that $a_n \leq Cn^c$ holds for all n , for some constants $C > 0$ and $c \in (1, 2)$. Lemma 3 then gives $x_k \gg k^{c/(c-1)}$.

Fix $X \geq 1$ and split the partition identity from Lemma 2:

$$X = \sum_{k \geq 2} |D_k \cap [1, X]| = \underbrace{\sum_{2 \leq k \leq K} |D_k \cap [1, X]|}_{(I)} + \underbrace{\sum_{k > K} |D_k \cap [1, X]|}_{(II)}, \quad (1)$$

where K will be chosen as $K = \lfloor X^{(c-1)/c} \rfloor$.

Bound on (I). Trivially $|D_k| \leq k - 1$, so

$$(I) \leq \sum_{k \leq K} (k - 1) \leq K^2/2 \ll X^{2-2/c}.$$

Bound on (II). For $k > K$ we have $x_k \gg k^{c/(c-1)} \gg K^{c/(c-1)} \asymp X$, so $x_k \gg X$ and in particular $x_k > 2X$ for all sufficiently large X (absorb a constant into the choice of K). The set

$$\{x_i : x_i \in [x_k - X, x_k]\}$$

is in bijection with $D_k \cap [1, X]$. Any two consecutive elements $x_{j-1} < x_j$ in $[x_k - X, x_k]$ satisfy

$$x_j - x_{j-1} \geq (x_j/C)^{1/c} \geq ((x_k - X)/C)^{1/c} \gg x_k^{1/c}$$

by Lemma 3 and $x_k - X \geq x_k/2$. Therefore

$$|D_k \cap [1, X]| \ll \frac{X}{x_k^{1/c}} \ll X \cdot k^{-1/(c-1)},$$

again using $x_k \gg k^{c/(c-1)}$. Summing,

$$(II) \ll X \sum_{k > K} k^{-1/(c-1)}.$$

Since $c < 2$, we have $1/(c-1) > 1$, so the sum converges; tail estimation gives

$$\sum_{k > K} k^{-1/(c-1)} \ll K^{1-1/(c-1)} = K^{(c-2)/(c-1)}.$$

Substituting $K = X^{(c-1)/c}$,

$$(II) \ll X \cdot X^{(c-1)/c \cdot (c-2)/(c-1)} = X^{1+(c-2)/c} = X^{2-2/c}.$$

Combining. From (1), $X \ll X^{2-2/c}$, i.e. $X^{2/c-1} \ll 1$. Since $c < 2$, the exponent $2/c - 1 > 0$, so the left-hand side tends to infinity as $X \rightarrow \infty$ — a contradiction. $\square$

4.3 Logarithmic refinement

The inequality $X \ll X^{2-2/c}$ is sharp at $c = 2$, and the contradiction fails there. By tracking the constants more carefully under a hypothesis $a_n \leq C n^2/f(n)$ (instead of $a_n \leq C n^c$), the same scheme yields the following.

Theorem 4 (Logarithmic refinement, after Bloom). *Let $f: \mathbb{N} \rightarrow \mathbb{R}_{>0}$ be nondecreasing, slowly varying (i.e. $f(\lambda n)/f(n) \rightarrow 1$ as $n \rightarrow \infty$ for every fixed $\lambda > 0$), unbounded, and satisfy*

$$\sum_{n \geq 2} \frac{1}{n f(n)} < \infty.$$

Then for any perfect difference set,

$$a_n \gg \frac{n^2}{f(n)} \quad \text{for infinitely many } n.$$

In particular, $a_n \gg n^2/(\log n)^{1+\varepsilon}$ infinitely often for every $\varepsilon > 0$, and $a_n \gg (n/\log n)^2$ infinitely often.

Proof. Suppose, for contradiction, that there exist $C > 0$ and $n_0 \in \mathbb{N}$ with

$$a_n \leq C n^2/f(n) \quad \text{for all } n \geq n_0. \quad (2)$$

Step 1 (gap bound). By PDS uniqueness, $a_{d_k} = x_k$ where $d_k = x_k - x_{k-1}$. For $d_k \geq n_0$, applying (2) gives $x_k \leq C d_k^2/f(d_k)$, equivalently

$$d_k \geq (x_k f(d_k)/C)^{1/2}. \quad (3)$$

Step 2 (size of x_k). We claim there exist $c_1 > 0$ and $k_0 \in \mathbb{N}$ such that

$$x_k \geq c_1 k^2 f(k) \quad (k \geq k_0). \quad (4)$$

Indeed, by (3) and $d_k \leq x_k$, slow variation of f gives, for k sufficiently large, $f(d_k) \geq (1/2)f(\sqrt{x_k})$, hence $d_k \geq c_2 \sqrt{x_k f(\sqrt{x_k})}$ with $c_2 = (2C)^{-1/2}$. Telescoping,

$$x_k \geq \sum_{i=k_0+1}^k d_i \geq c_2 \sum_{i=k_0+1}^k \sqrt{x_i f(\sqrt{x_i})}.$$

This recurrence is monotone in x_i ; comparing with the differential equation $dy/dk = c_2 \sqrt{y f(\sqrt{y})}$ (separable as $dy/\sqrt{y} = c_2 \sqrt{f(\sqrt{y})} dk$), integrating, and using slow variation $f(\sqrt{y}) \asymp f(y^{1/4}) \asymp \dots$ to absorb the self-dependence into a constant, we obtain $\sqrt{x_k} \geq (c_2/2) k \sqrt{f(k)} (1 - o(1))$, hence (4) with $c_1 = c_2^2/8$.

Step 3 (anchor window). Fix X large. Let $K = K(X)$ be the largest integer with $x_K \leq 2X$. By (4), $K^2 f(K) \leq 2X/c_1$, so

$$K \leq \sqrt{2X/(c_1 f(K))} \leq \sqrt{2X/(c_1 f(\sqrt{X/f(\sqrt{X})}))} \sim \sqrt{2X/(c_1 f(\sqrt{X}))}, \quad (5)$$

where the last step uses f slowly varying (since $K = \Theta(\sqrt{X/f(\sqrt{X})})$ gives $f(K) \asymp f(\sqrt{X})$). Crucially, since $f(\sqrt{X}) \rightarrow \infty$,

$$K^2 = o(X) \quad (X \rightarrow \infty). \quad (6)$$

For $k > K$ we have $x_k > 2X$. The window $W_k := [x_k - X, x_k]$ contains some subset of A . For any two consecutive $x_{j-1} < x_j$ in W_k , (3) gives $d_j \geq \sqrt{x_j f(d_j)/C}$. Now $x_j \geq x_k - X \geq x_k/2$, so $\sqrt{x_j} \geq \sqrt{x_k}/\sqrt{2}$; slow variation of f on $[d_j, x_j]$ gives $f(d_j) \geq (1/2)f(\sqrt{x_k})$ for k large. Therefore

$$d_j \geq (1/\sqrt{4C}) \sqrt{x_k f(\sqrt{x_k})}.$$

By (4) and slow variation, $\sqrt{x_k f(\sqrt{x_k})} \geq c_3 k f(k)$ for k large, with $c_3 > 0$ depending only on c_1 . Hence

$$|D_k \cap [1, X]| = |A \cap W_k| \leq \frac{X}{c_3 k f(k)} + 1. \quad (7)$$

Step 4 (partition sum). By Lemma 2,

$$X = \sum_k |D_k \cap [1, X]|.$$

Split at K :

$$X = \underbrace{\sum_{2 \leq k \leq K} |D_k \cap [1, X]|}_{(I)} + \underbrace{\sum_{k > K} |D_k \cap [1, X]|}_{(II)}.$$

For (I), trivially $|D_k| \leq k - 1$, so $(I) \leq K(K - 1)/2 \leq K^2/2 = o(X)$ by (6).

For (II), apply (7):

$$(II) \leq \frac{X}{c_3} \sum_{k > K} \frac{1}{k f(k)} + M_K,$$

where $M_K = |\{k > K : |A \cap W_k| \geq 1\}|$ counts which large- k windows are non-empty. Each non-empty W_k contributes a distinct $x \in A$ (since W_k ranges over disjoint anchor positions x_k), and any two consecutive x_k, x_{k+1} in this counting differ by $d_{k+1} \geq c_4 \sqrt{x_k f(\sqrt{x_k})} \geq c_4 \sqrt{X}$ (using $x_k > 2X$ and slow variation). So $M_K \leq c_5 \sqrt{X}$ (the number of A -anchors that could possibly have a window intersecting $[1, X]$ is bounded by Sidon density of A). In particular $M_K = O(\sqrt{X}) = o(X)$.

Let $\varepsilon_K := \sum_{k > K} 1/(k f(k))$. By the convergence hypothesis, $\varepsilon_K \rightarrow 0$ as $K \rightarrow \infty$, and in particular $\varepsilon_K \rightarrow 0$ as $X \rightarrow \infty$. Thus

$$(II) \leq \frac{X \varepsilon_K}{c_3} + o(X).$$

Step 5 (contradiction). Combining,

$$X \leq \frac{K^2}{2} + \frac{X \varepsilon_K}{c_3} + o(X) = \frac{X \varepsilon_K}{c_3} + o(X).$$

For X large, $\varepsilon_K < c_3/2$ and the $o(X)$ term is less than $X/4$, so

$$X \leq \frac{X}{2} + \frac{X}{4} = \frac{3X}{4},$$

which is false. This contradicts (2).

Hence (2) fails for every C : for each $C > 0$, there are infinitely many n with $a_n > C n^2/f(n)$, i.e. $a_n \gg n^2/f(n)$ infinitely often. $\square$

The case $f(n) = (\log n)^{1+\varepsilon}$ ($\varepsilon > 0$) gives $a_n \gg n^2/(\log n)^{1+\varepsilon}$ infinitely often; taking $\varepsilon \rightarrow 0^+$ recovers $a_n \gg n^{2-o(1)}$ from Theorem 3. The convergence boundary $\sum 1/(nf(n)) < \infty$ is sharp *for this argument*: at $f(n) = \log n$ (where the sum diverges, $\sum 1/(n \log n) = \infty$), the tail ε_K in Step 4 no longer goes to zero with K and the contradiction collapses. Pushing past this boundary would require either a sharper density estimate on the anchor window in Step 3 or a non-partition counting identity.

[*Typographical note on the live problem page*] At time of writing (April 30, 2026), the problem page erdosproblems.com/1194 states the conclusion of Theorem 4 as “for any function f such that $\sum 1/(nf(n))$ *diverges*.” This appears to be a copy-edit error: *convergence* of $\sum 1/(nf(n))$ is the natural sufficient condition for the proof, and is what Bloom states (with example $f(n) = (\log n)^2$) in the comment thread after natso26’s correction (post 5782). With the divergence condition, $f(n) = 1$ would qualify and imply $a_n \gg n^2$ infinitely often, which the proof does *not* establish.

5 What is *not* given by these arguments

Theorems 2 and 3 are *infinitely often* statements: they exhibit a sparse subsequence $n_j \rightarrow \infty$ on which a_{n_j} is large. They place no constraint on a_n for “most” n .

Empirically (see `verify_and_analyze.py`), the running minimum $\inf_{n \leq N} a_n/n^2$ tends to zero in the greedy PDS: $\min_{n \leq 1898} a_n/n^2 \approx 5.6 \times 10^{-4}$. So a “for-all-large- n ” version of $a_n \gg n^2$ is empirically false even for the greedy PDS, and the original Erdős question is best reformulated as a statement on the running maximum

$$M(N) := \max_{n \leq N} a_n.$$

Theorems 2 and 3 translate into $M(N) \gg N \log N$ and $M(N) \gg N^{2-o(1)}$ respectively. Lev’s greedy upper bound is $M(N) = O(N^3)$, refined empirically to $M(N) \approx 0.0293 N^3$ for $N \in [1500, 1898]$.

The unresolved question is therefore: *is there a constant $\alpha \in (2, 3]$ such that $M(N) \asymp N^\alpha$? In particular, can the upper bound be brought down from N^3 , or the lower bound brought up to match?* The Cilleruelo–Nathanson method [3] for transferring results from dense Sidon sets to PDS is a candidate route on the upper side.

References

- [1] H. Halberstam and K. F. Roth, *Sequences*, Oxford University Press, 1966.
- [2] V. F. Lev, *Reconstructing integer sets from their representation functions*, Electron. J. Combin. 11 (2004), R78.
- [3] J. Cilleruelo and M. B. Nathanson, *Perfect difference sets constructed from Sidon sets*, Combinatorica 28 (2008), 401–414.